Title

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- facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024

Identifying Ownership Changes: Why Ownership Is Non-Trivial

- Ownership is often **layered** (organizations owning organizations)
- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Entity Type	Owner Name	Share
Direct ownership			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
Indirect ownership			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Example (Legal Restructuring)

- HCRIS ownership change flags may reflect **purely legal restructuring**
- NHC ownership data allows us to verify whether economic control changes
- **Example from the data:**
  - *Before:*
    - ESTES JAMES (84%)
    - BURCHFIELD JOHN (10%)
    - LEE CLAUDE (6%)
  - *After:*
    - JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
    - JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- We do **not** treat this as a meaningful ownership change

Identifying Ownership Changes: Definition and Sample

- **Step 1 (flag):** HCRIS ownership change flag identifies candidate change events
- **Step 2 (validate & date):** NHC ownership panel identifies whether control changes and when
- **Ownership change criteria (NHC):**
  - An ownership change occurs between $t-1$ and t when **majority control (50%)** transfers
  - Partial / missing shares are normalized; missing ownership shares are assigned equal weights within a level
  - We do not flag changes driven by name changes or within-family reclassification (e.g., trust/LLC relabeling)
- **Final sample restriction):**
  - Facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
  - Change month is the first month NHC indicates a change

Ownership Change Agreement: NHC vs HCRIS

Cross-tab of Ownership Changes: NHC vs HCRIS

Count of changes in NHC \ Count of changes in HCRIS	0	1	2+
2+	1,137	1,377	535
1	1,874	1,589	101
0	6,217	499	15

Dependent Variables: Nurse Staffing (PBJ)

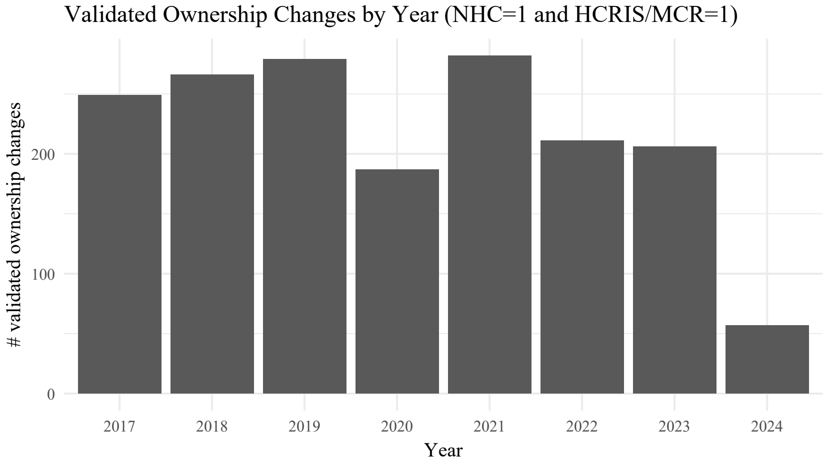
- We study four staffing measures:
  - **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
  - **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks
  - **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
  - **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

PBJ Cleaning and Final Sample

- We apply CMS technical guidance to remove implausible staffing reports:
  1. RN HPPD $= 0$ **and** LPN HPPD $= 0$
  2. Total HPPD < 1.5 or > 12
  3. CNA HPPD > 5.25
- Additional restrictions:
  - Drop facilities with beds < 15
  - Drop hospital-based facilities
- **Final sample:**
  - 7,806 facilities
  - 632,231 facility-month observations

Validated Ownership Changes Over Time (Yearly)



Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD event study around ownership transitions
- Identification comes from **within-facility changes** relative to facilities not yet treated

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \mathbf{1}\{event\_time_{it} = \tau\} \cdot Treat_i + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (1)$$

- Y_{it}: staffing outcome measured in **HPPD**
- α_i facility FE, λ_t month FE; X_{it}' controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- Standard errors clustered at the facility level

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- **Primary threat: anticipation / timing mismatch**
  - Staffing may adjust before the recorded transition month.

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- Administrative "event dates" may reflect reporting/legal dates, not the first operational change
- **Implication:** months immediately before $\tau = 0$ may violate parallel trends

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- We use $\tau = -4$ as the reference period
- Post change coefficients compare to a pre-period less likely to be influenced by transition activity

Assessing Parallel Trends

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- **Takeaway:**
  - With the full pre-window (including $\tau = -3, -2, -1$), pre-trends show evidence of violation.
  - Excluding anticipatory months (donut) yields pre-trends more consistent with parallel trends.

Parallel Trends Tests (Joint Wald)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Levels (HPPD)				
2 Year Full Pre-Window	107.49 (23) [0.0000]	41.63 (23) [0.0100]	46.13 (23) [0.0029]	51.22 (23) [0.0006]
2 Year Window with Donut	34.83 (20) [0.0210]	36.57 (20) [0.0132]	17.83 (20) [0.5984]	24.92 (20) [0.2046]
1 Year Window with Donut	7.55 (8) [0.4788]	10.49 (8) [0.2322]	3.89 (8) [0.8666]	6.08 (8) [0.6387]
Panel B: Logs (HPPD)				
2 Year Full Pre-Window	78.21 (23) [0.0000]	45.72 (23) [0.0032]	34.54 (23) [0.0578]	49.22 (23) [0.0012]
2 Year Window with Donut	17.14 (20) [0.6436]	37.42 (20) [0.0104]	13.28 (20) [0.8651]	22.26 (20) [0.3268]
1 Year Window with Donut	3.33 (8) [0.9122]	13.48 (8) [0.0965]	7.50 (8) [0.4837]	6.33 (8) [0.6102]

Notes: Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report χ^2 (df) [p-value].
Donut excludes $\tau \in \{-3, -2, -1\}$ and uses $\tau = -4$ as the reference period.

Summary Effect: TWFE Difference-in-Differences

$$Y_{it} = \beta\, Post_{it} + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- $Post_{it} = 0$ before an ownership change and $= 1$ in all months after a change
- β summarizes the average post-change staffing effect
- Quantifies average changes for cleaner and relevant results